Patch-Based Tokenisation of Discrete-Time Signals

Coursework — Points 1 & 2

Contents

1 Point 1 — Patch-Based Tokenisation as an Operator; Conditions for Identical Patches

1.1 Setup and Formal Definition

Let $x[n]$ be a discrete-time signal with $n \in \mathbb{Z}$, sampled at frequency f_s. Patch-based tokenisation extracts a sequence of tokens by sliding a window of size P samples with stride S:

$$\mathbf{p}_k = \big(x[kS],\ x[kS+1],\ \dots,\ x[kS+P-1]\big) \in \mathbb{R}^P, \qquad k = 0, 1, 2, \dots \tag{1}$$

Definition 1 (Patch Tokenisation Operator)**.** *The map $\mathcal{T} : \ell^\infty(\mathbb{Z}) \to (\mathbb{R}^P)^{\mathbb{N}}$ defined by (1) is called the* patch tokenisation operator*. It is linear: collecting K patches yields*

$$\mathbf{P} = \mathcal{T}\{x\} = \mathbf{W} \cdot \mathbf{x}, \tag{2}$$

where $\mathbf{W} \in \{0,1\}^{KP \times N}$ is a binary selection matrix whose k-th block of P rows picks the P consecutive samples starting at index kS.

1.2 Condition for Two Consecutive Patches to Be Identical

Proposition 1. *Patches $\mathbf{p}_k$ and $\mathbf{p}_{k+1}$ are identical if and only if*

$$x[n] = x[n+S] \quad \forall\, n \in \{kS,\, kS+1,\, \dots,\, kS+P-1\}. \tag{3}$$

Globally, if the signal is periodic with period T_0 (in samples), identical consecutive patches occur whenever

$$\boxed{S = m \cdot T_0, \qquad m \in \mathbb{Z}^+.} \tag{4}$$

Equivalently, in terms of signal frequency f and sampling frequency f_s:

$$f = m \cdot \frac{f_s}{S}, \qquad m \in \mathbb{Z}^+. \tag{5}$$

Proof. $\mathbf{p}_k = \mathbf{p}_{k+1}$ component-wise requires $x[kS+j] = x[(k+1)S+j]$ for all $j \in \{0, \dots, P-1\}$, i.e. $x[n] = x[n+S]$ on that window. For a periodic signal with $T_0 = f_s/f$ samples, this holds whenever S is an integer multiple of T_0, giving (4). □

1.3 Loss of Observability

When $\mathbf{p}_k = \mathbf{p}_{k+1}$ for all k, the token sequence is constant — the model sees the same vector repeated. The *rank* of the token matrix $\mathbf{P}$ collapses to 1. Consequently:

- Temporal variation is invisible to the model.
- Any two signals differing only by a phase shift of S samples produce the same token sequence: they are **observationally equivalent**.
- Even when patches are not fully identical but merely highly correlated ($S \approx mT_0$), the effective information in the token sequence is reduced (*soft* loss of observability).

2 Point 2 — Relationship Between f_s, P, S, f; Phase-Locking and Degeneracies

2.1 The Patch-Induced Frequency

The tokeniser introduces a characteristic *patch frequency*:

$$f_{\text{patch}} = \frac{f_s}{S}. \tag{6}$$

This is the rate at which new patches are produced (in Hz, referred to original time), analogous to a resampling rate in the token domain.

2.2 Phase-Locking Condition

Consider a sinusoidal signal $x[n] = A\cos\big(2\pi f n/f_s + \phi\big)$.
Definition 2 (Phase-Locking). *The signal* phase-locks *to the patch grid when its period $T_0 = f_s/f$ (in samples) is* commensurate *with S:*

$$\frac{S \cdot f}{f_s} \in \mathbb{Q}. \tag{7}$$

The degenerate (critical) case occurs when this ratio is an integer:

$$f = m \cdot \frac{f_s}{S} = m \cdot f_{\text{patch}}, \qquad m \in \mathbb{Z}^+, \tag{8}$$

*i.e. f is a **harmonic of** f_{patch}.*

2.3 Effect of Patch Size P

A single patch of P samples spans a duration P/f_s seconds and covers $P \cdot f/f_s$ cycles of the signal. When

$$\frac{P \cdot f}{f_s} \in \mathbb{Z}^+, \quad \text{i.e.} \quad f = k \cdot \frac{f_s}{P}, \quad k \in \mathbb{Z}^+, \tag{9}$$

the patch contains an exact integer number of cycles: its content is *internally redundant.*

The most severe degeneracy occurs when both the stride and the patch-size conditions hold simultaneously:

$$f = \frac{\ell \cdot f_s}{\operatorname{lcm}(P,\, S)}, \qquad \ell \in \mathbb{Z}^+. \tag{10}$$

2.4 Token-Space Degeneracies

At $f = m\, f_s/S$ (harmonics of f_{patch}).

- All patches are translates of the same waveform by a phase that is a multiple of 2π: they become **identical** (cf. Point 1).
- The token sequence is constant; it encodes **no phase information**.
- Two signals at the same frequency but different phases (shifted by any multiple of S samples) map to the same token sequence: *token-space invariance.*

At $f = k\,f_s/P$ (but not harmonics of f_{patch}).

- Each patch is internally periodic, so its content is *within-patch redundant.*
- Inter-patch variation still exists (partial degeneracy).

2.5 Overlap, Stride, and Blind-Spot Frequencies

When $S < P$ (overlapping patches), consecutive patches share $P - S$ samples, partially mitigating the aliasing effect even near harmonic frequencies. For $S \geq P$ (no overlap or gaps), the aliasing is maximal; the **blind-spot frequencies** are:

$$\boxed{f_{\text{blind}} = m \cdot \frac{f_s}{S}, \qquad m = 1, 2, 3, \ldots, \left\lfloor \frac{S}{2} \right\rfloor.} \tag{11}$$

The upper limit follows from the Nyquist criterion applied to the token sequence, whose effective sample rate is f_s/S. This consegunces followed what expressed in [1]

2.6 Summary

Condition	Effect
$f = m\,f_s/S$	Full phase-locking $\rightarrow$ identical patches $\rightarrow$ constant token sequence
$f = k\,f_s/P$	Internal patch periodicity $\rightarrow$ within-patch redundancy
Both simultaneously	Maximal degeneracy
$S < P$ (overlap)	Partial mitigation; residual phase information
$S \geq P$ (no overlap)	Full degeneracy at blind-spot frequencies

Table 1: Summary of conditions and their effects on token-space observability.

References

[1] Alessandro Pagani, Marco Cominelli, Liying Han, Gaofeng Dong, Sergio Benini, Francesco Gringoli, Mattia Savardi, Mani B. Srivastava, Trevor Bihl, Erik P. Blasch, Daniel O. Brigham, Kara Combs, Lance M. Kaplan, and Federico Cerutti. Preliminary Insights in Chronos Frequency Data Understanding and Reconstruction.